DTU

Model-based Machine Learning

Classification models

Outline

Case study: Modeling travel choices

Logistic regression

Hierarchical models

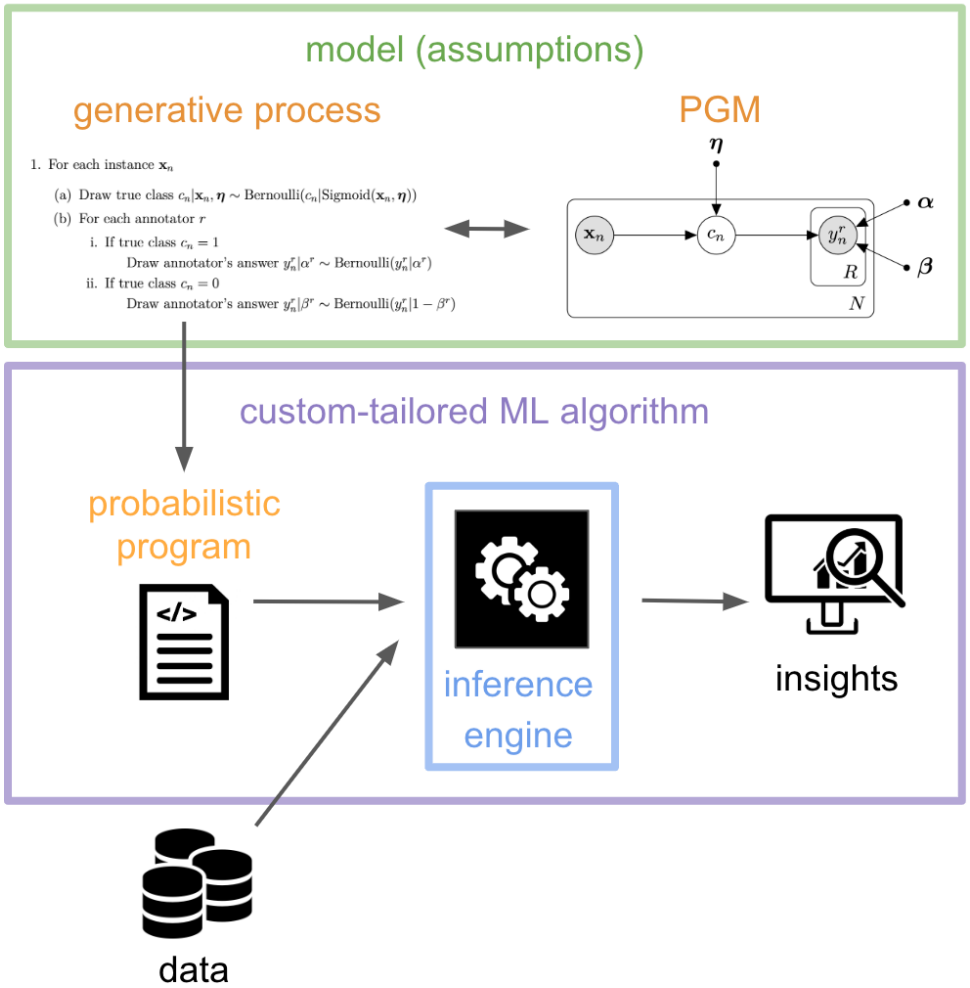
Special cases

Road to MBML: where are we?

Foundations of PGMs and probabilistic programming
Probabilistic modelling in various contexts - your modelling toolbox
Bayesian inference (exact and approximate)

Week	Topic
1	Intro to the course + Prob. review
2	PGM foundations
3	PGM foundations II
4	Freq. vs Bayesian + Prob. Prog. + Mixture models
5	Regression models
6	Classification and Hierarchical models
7	Temporal models
8	Topic Models
9	Markov-chain Monte Carlo (MCMC)
10	Variational inference
11	Generative models
12	Gaussian processes
13	Project support

Model-based Machine Learning



Learning objectives

At the end of this lecture, you should be able to:

- Explain what (Bayesian) logistic regression is and explain its underlying assumptions
- Explain the extension of logistic regression to multiple classes
- Explain the underlying concepts and assumptions behind hierarchical models
- Relate different ways of modelling the dependency of a discrete random variable on other variables and justify their suitability for a problem
- Relate logistic regression to choice models and recognise the identification requirements of choice models
- Recognise ways to analyse ordinal data.
- Implement the modelling techniques above in Pyro

Case study: Modeling travel mode choices

- Travel diary data
  - 394 survey observations from 80 individuals
  - 4 travel modes: plane, train, bus or car
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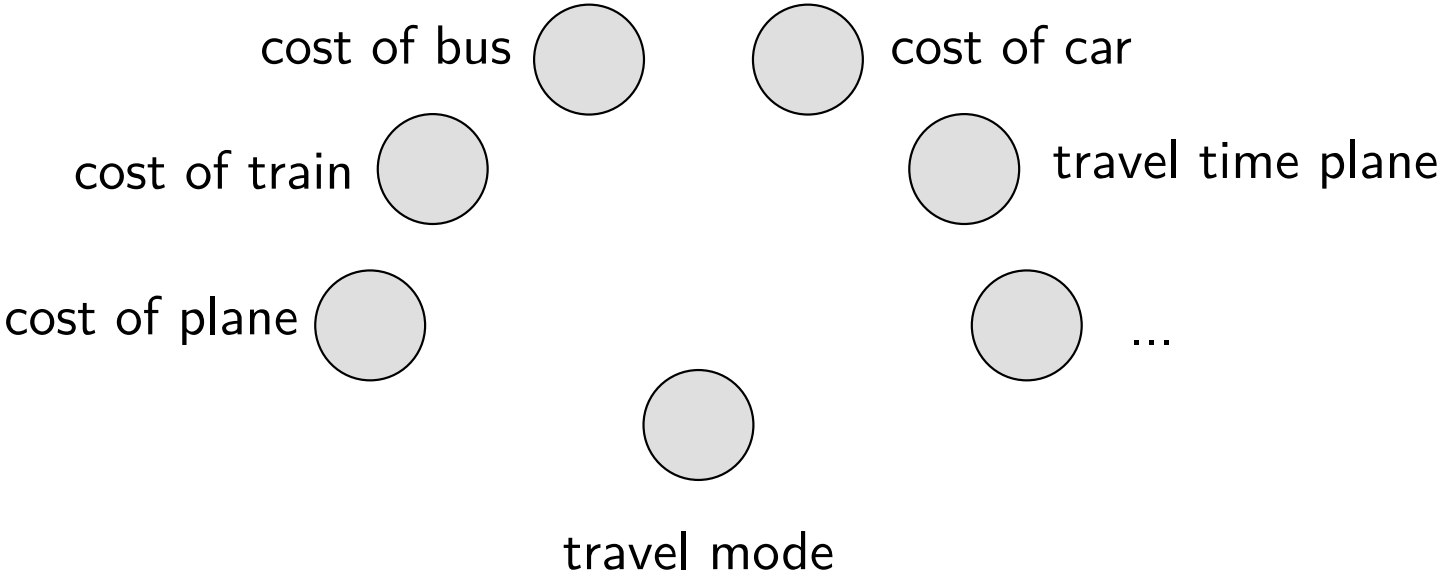
  - Household income
  - Traveling group size
- Some possible applications:
  - Understanding people's choices
  - Developing pricing policies
  - Incentivising mode change
  - Suggesting car pooling

Modeling travel mode choices (cont'd)

- Let's start thinking about the graphical model...



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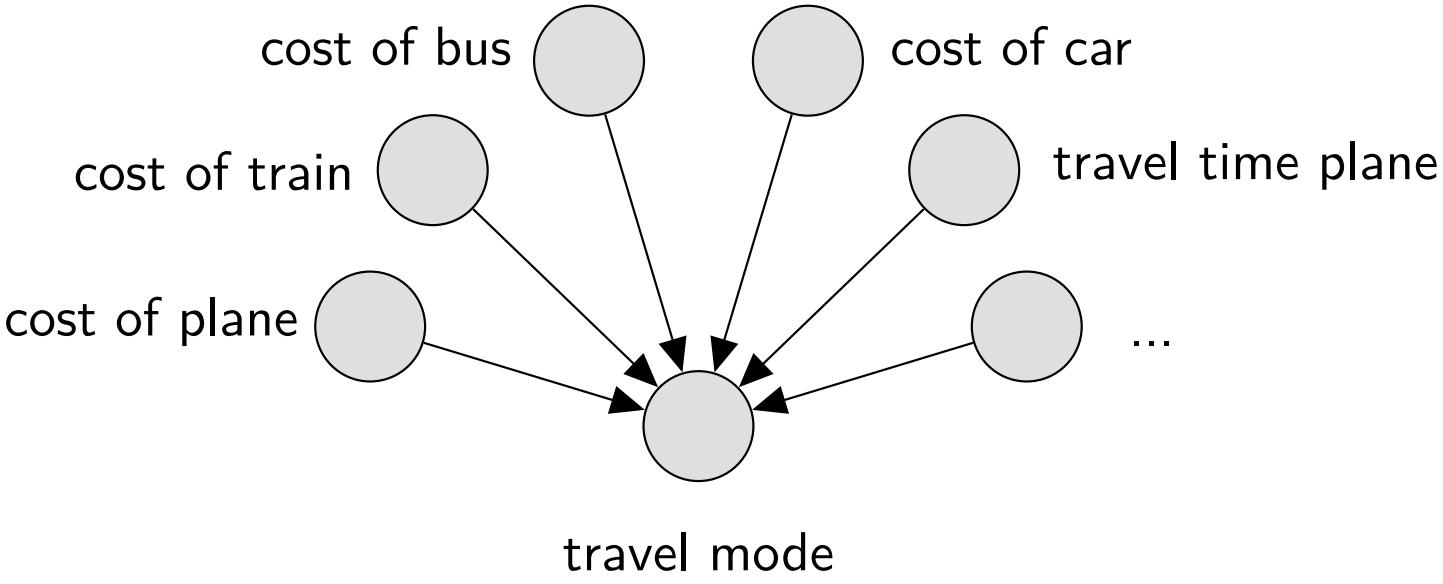

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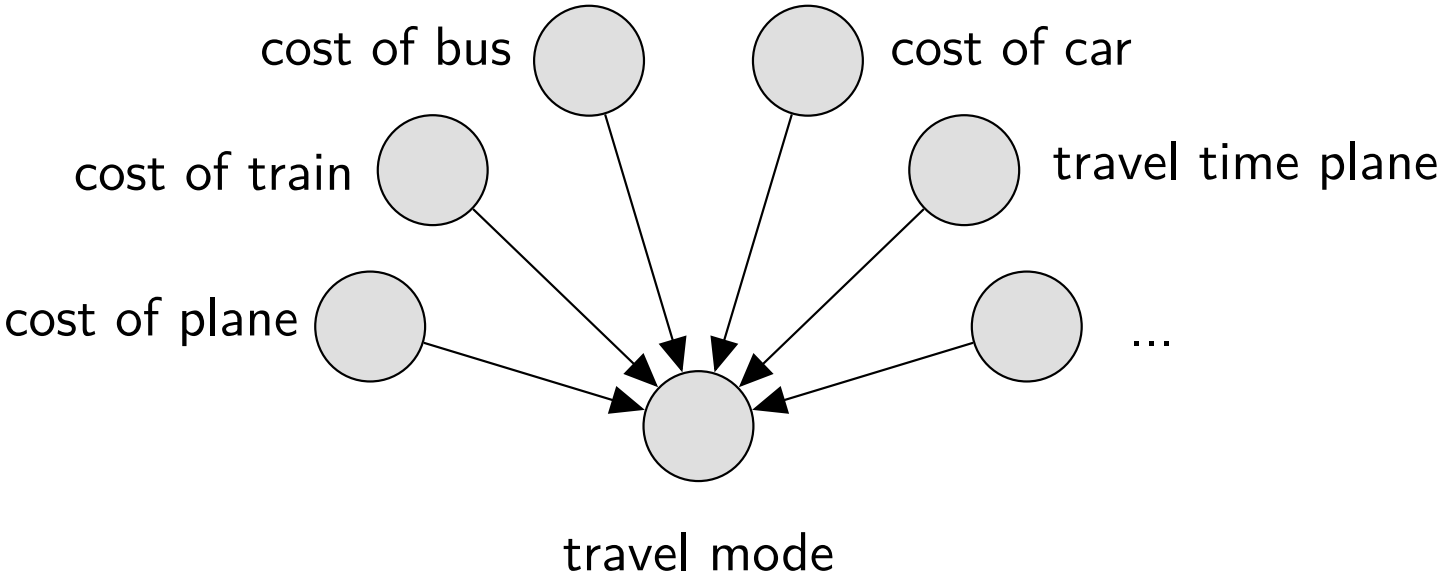
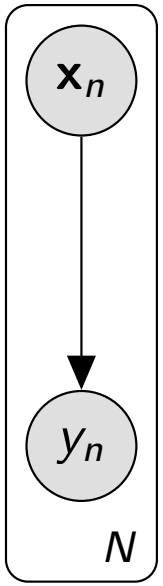


- What distribution should we assign to the "travel mode" variable?
  - Travel mode is a **discrete variable**!
  - We are now in a **classification** setting
- How should we model the dependency of the travel mode on the other variables?

Discrete output variables

- We can represent our model for the entire dataset compactly as:



N is the number of trips in the dataset
y_n is the travel mode of the nth trip in the dataset
$\mathbf{x}_n$ is a vector with $\{$cost of plane, cost of train, $\ldots\}$ for trip n

- Looks familiar?

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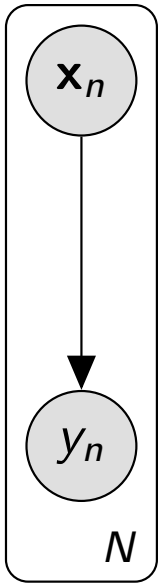

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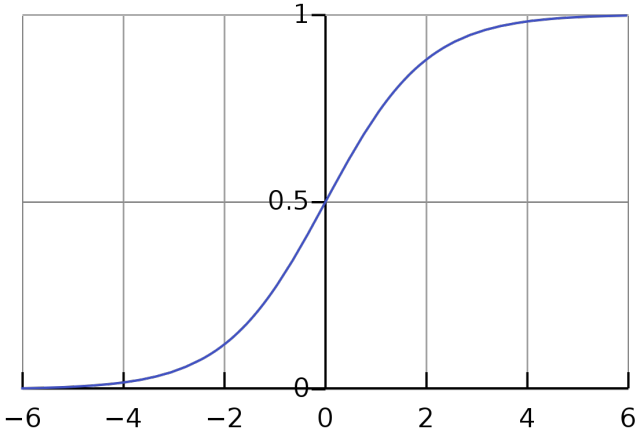
- Looks familiar?
- But how should we model the dependency of y_n on $\mathbf{x}_n$?
  - We can assume a parameterized linear relationship: $y_n = \beta^\mathsf{T}\mathbf{x}_n$
  - But $y_n \notin \mathbb{R}$! Instead: $y_n \in \{$plane, train, bus, car$\}$

Binary logistic regression

- Consider the binary case: $y_n \in \{0, 1\}$
- We need a function that maps from $\mathbb{R}$ to $[0, 1]$
- A sigmoid ("S"-shaped) function does precisely that!
- E.g. logistic sigmoid:

$$\begin{aligned}\text{Sigmoid}(z) &= \frac{1}{1 + e^{-z}} \\ &= \frac{e^z}{e^z + 1}\end{aligned}$$



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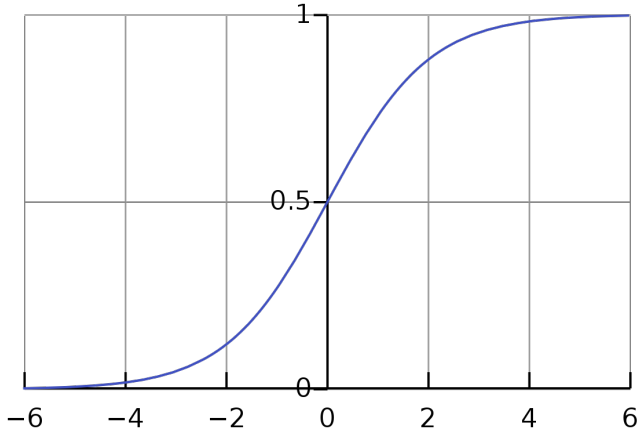

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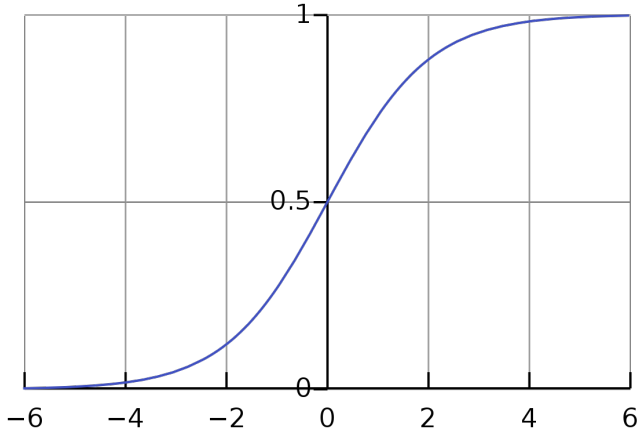

- We can define $z_n = \beta^\mathsf{T} \mathbf{x}_n$
- The value of $\text{Sigmoid}(z_n)$ can then be interpreted as the probability of the nth instance belonging to class "1": $p(y_n = 1)$
- The probability of class "0" is simply: $p(y_n = 0) = 1 - \text{Sigmoid}(z_n)$

Binary logistic regression as a graphical model

- We have a dataset $\mathcal{D}$ consisting of N observations of the targets $y_n \in \{0, 1\}$ which depend on their corresponding explanatory variables $\mathbf{x}_n$

$$\mathcal{D} = \{\mathbf{x}_n, y_n\}_{n=1}^{N}$$

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- Generative process
  1. Draw coefficients $\beta \sim \mathcal{N}(\beta|\mathbf{0}, \lambda\mathbf{I})$
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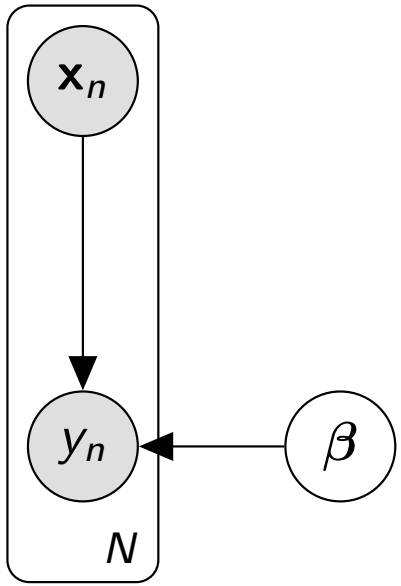

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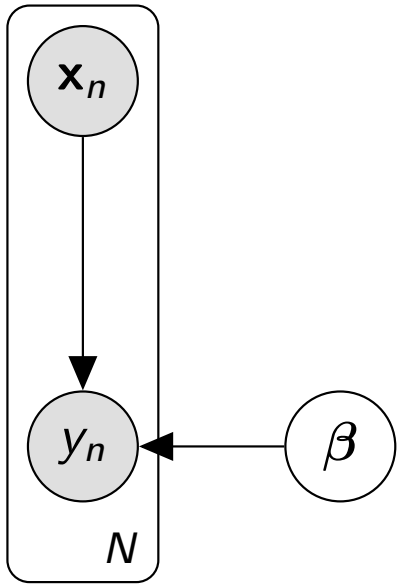

- Joint probability distribution factorizes as

$$p(\mathbf{y}, \beta|\mathbf{X}, \lambda) = \underbrace{p(\beta|\lambda)}_{\text{prior}} \times \underbrace{\prod_{n=1}^{N} p(y_n|\beta, \mathbf{x}_n)}_{\text{likelihood}}$$

where $\mathbf{y} = \{y_n\}_{n=1}^N$, $\mathbf{X} = \{\mathbf{x}_n\}_{n=1}^N$ and β are the model parameters.

Multi-class logistic regression

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- The generalization of the logistic sigmoid to multiple outputs is the **softmax**:

$$\text{Softmax}(\mathbf{x}_n, \boldsymbol{\beta}_1, \ldots, \boldsymbol{\beta}_C)_c = \frac{\exp(\boldsymbol{\beta}_c^\mathsf{T}\mathbf{x}_n)}{\sum_{k=1}^{C}\exp(\boldsymbol{\beta}_k^\mathsf{T}\mathbf{x}_n)}, \quad \text{for } c \in \{1, \ldots, C\}$$

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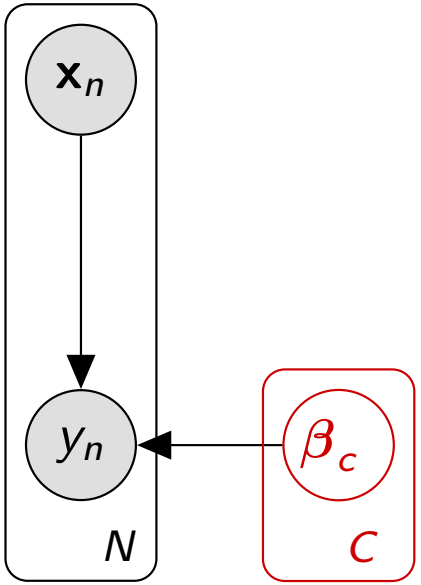
Note

In practice, we only need $C-1$ vectors of params (e.g., set $\boldsymbol{\beta}_C = \mathbf{0}$). Probabilities must sum to 1. This will ensure that your model implementation is **identifiable**!

Multi-class logistic regression as a graphical model

- Updated graphical model



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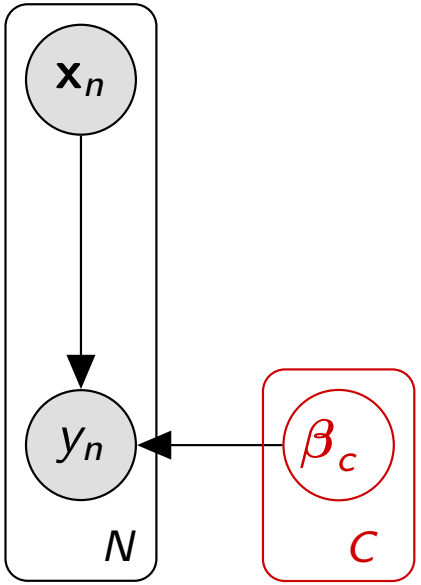

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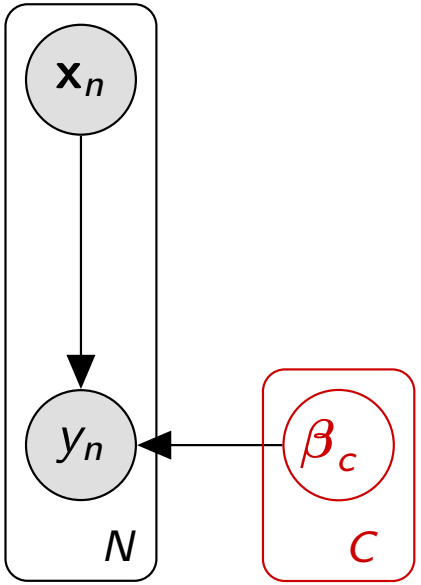

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- **Goal:** compute **posterior** distribution on $\beta_1, \ldots, \beta_C$
- Following Bayes' theorem

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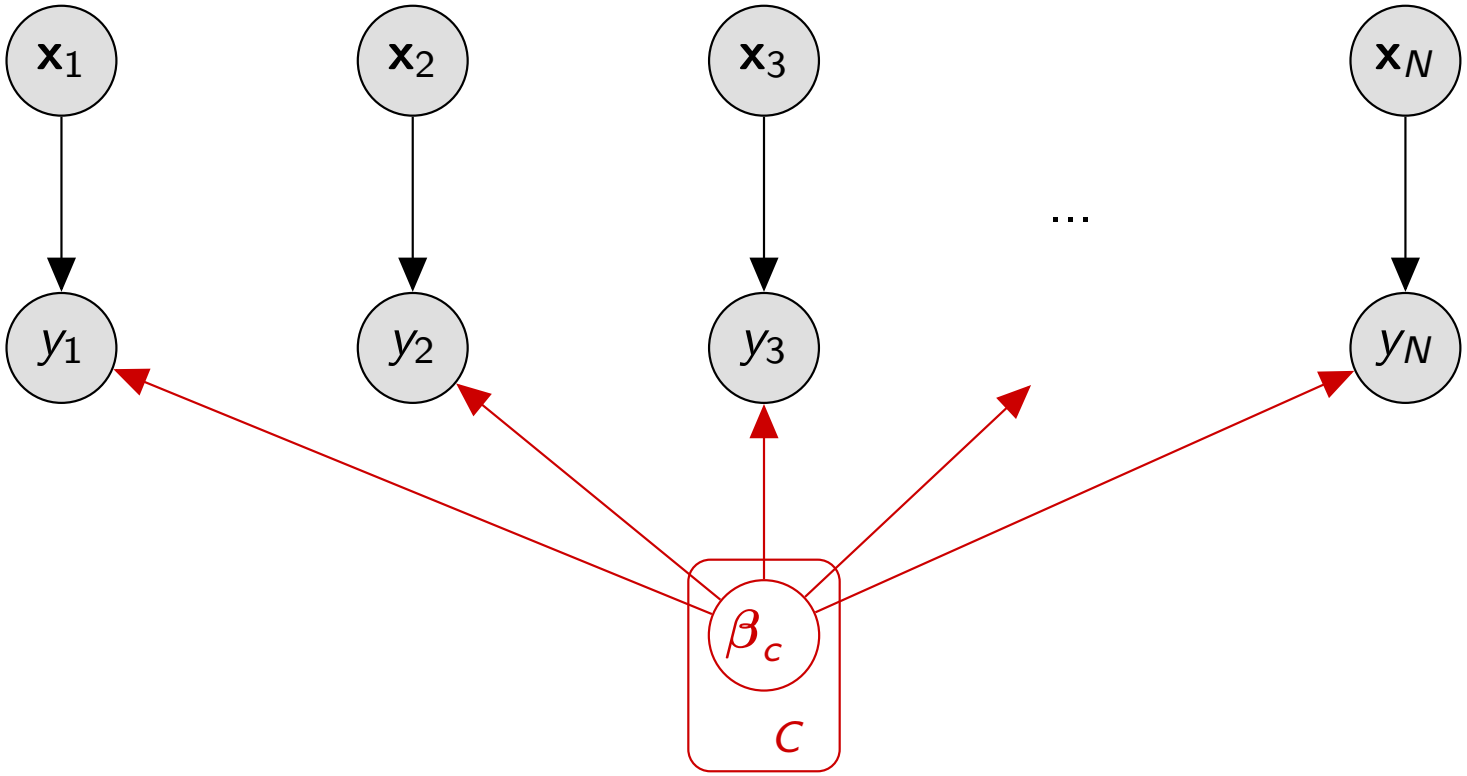
- Exact inference is intractable
- Must resort to **approximate inference** methods
- Not a problem for Pyro :-)

Playtime!

- Ancestral sampling from multi-class logistic regression model
  - See “05 - Classification models - Part 1.ipynb" notebook
  - Expected duration: 15 minutes
- Bayesian multi-class logistic regression model of travel mode choices
  - See “05 - Classification models - Part 2.ipynb" notebook
  - Expected duration: 1 hour

Going back to our travel mode choice case study...

- Let's revise the **modeling assumptions** that we made



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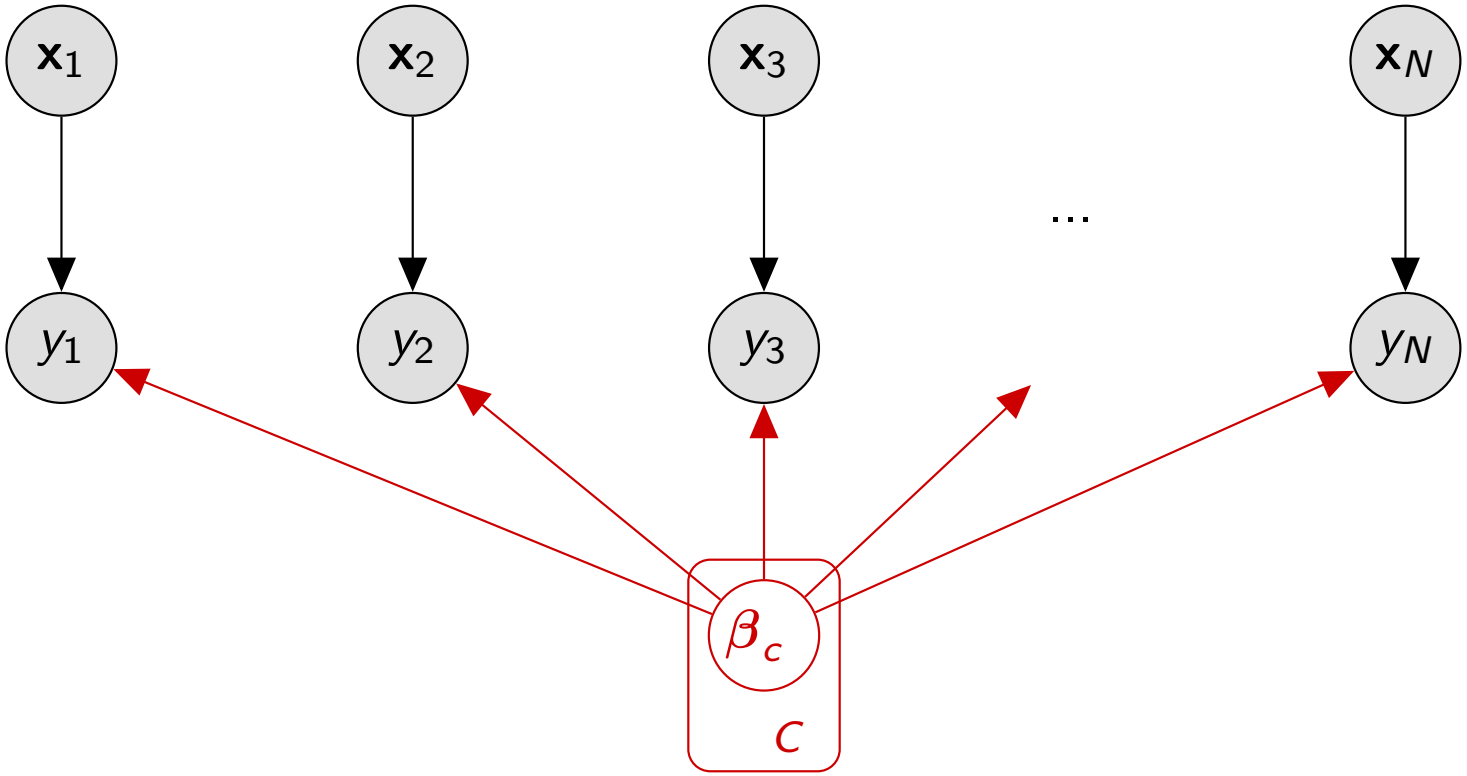
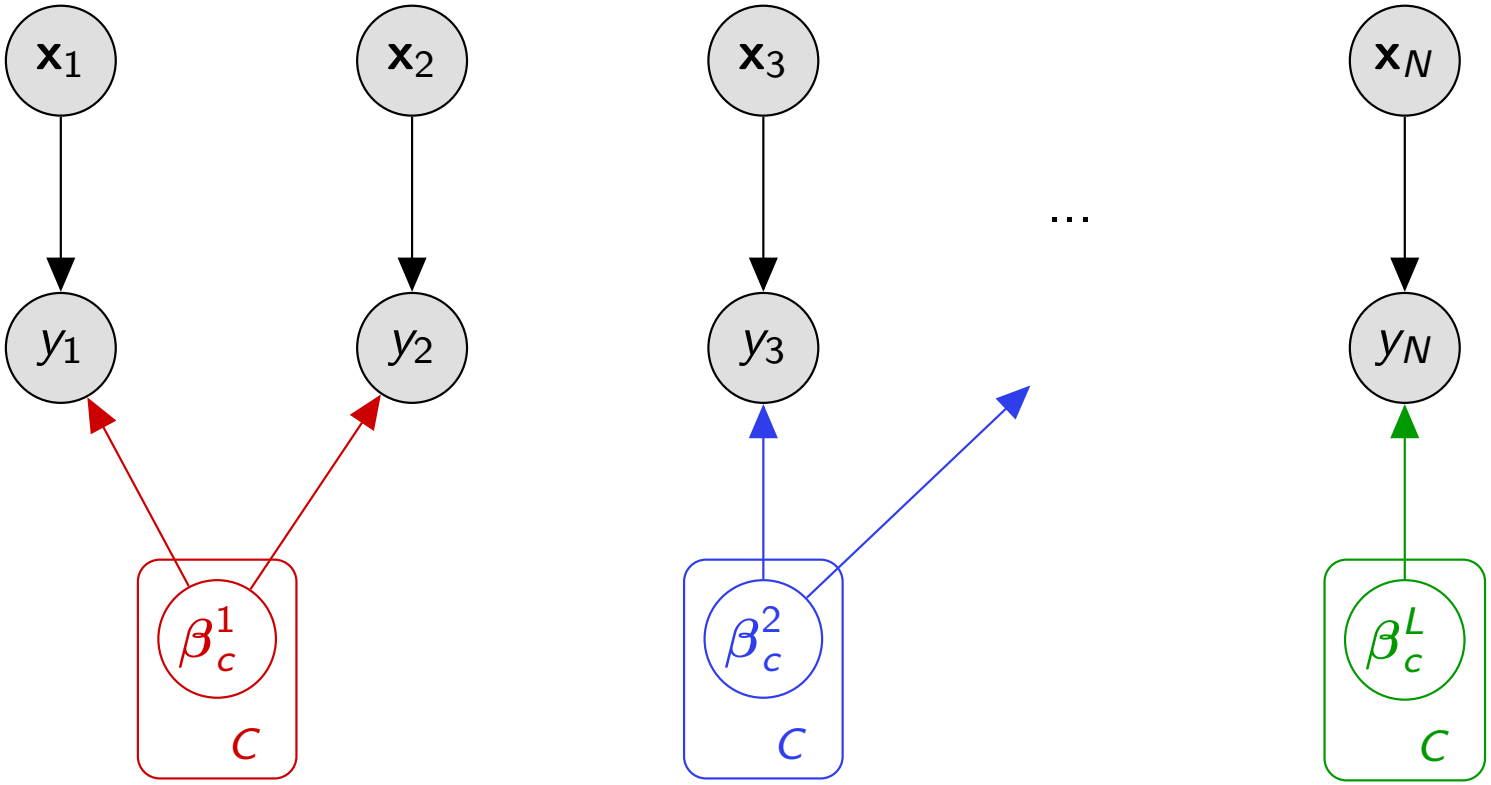


- Single set of parameters $\{\beta_1, \ldots, \beta_C\}$ for **all** the observations
  - This corresponds to saying that all individuals give the same importance (weight) to all the features (e.g. travel time) and have the same biases!

Going back to our travel mode choice case study...

- Alternatively, we can assign each individual his/her own parameters



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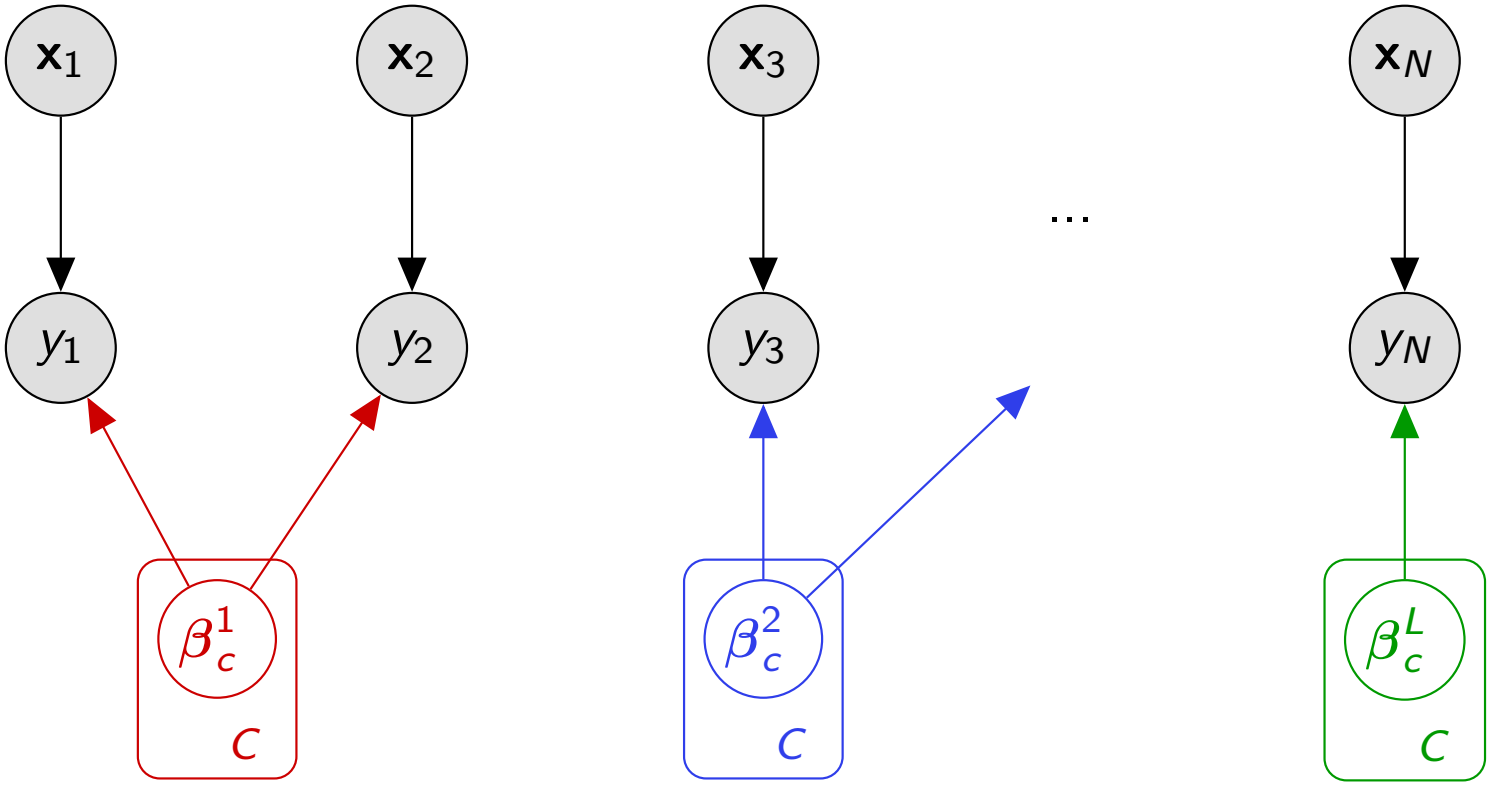
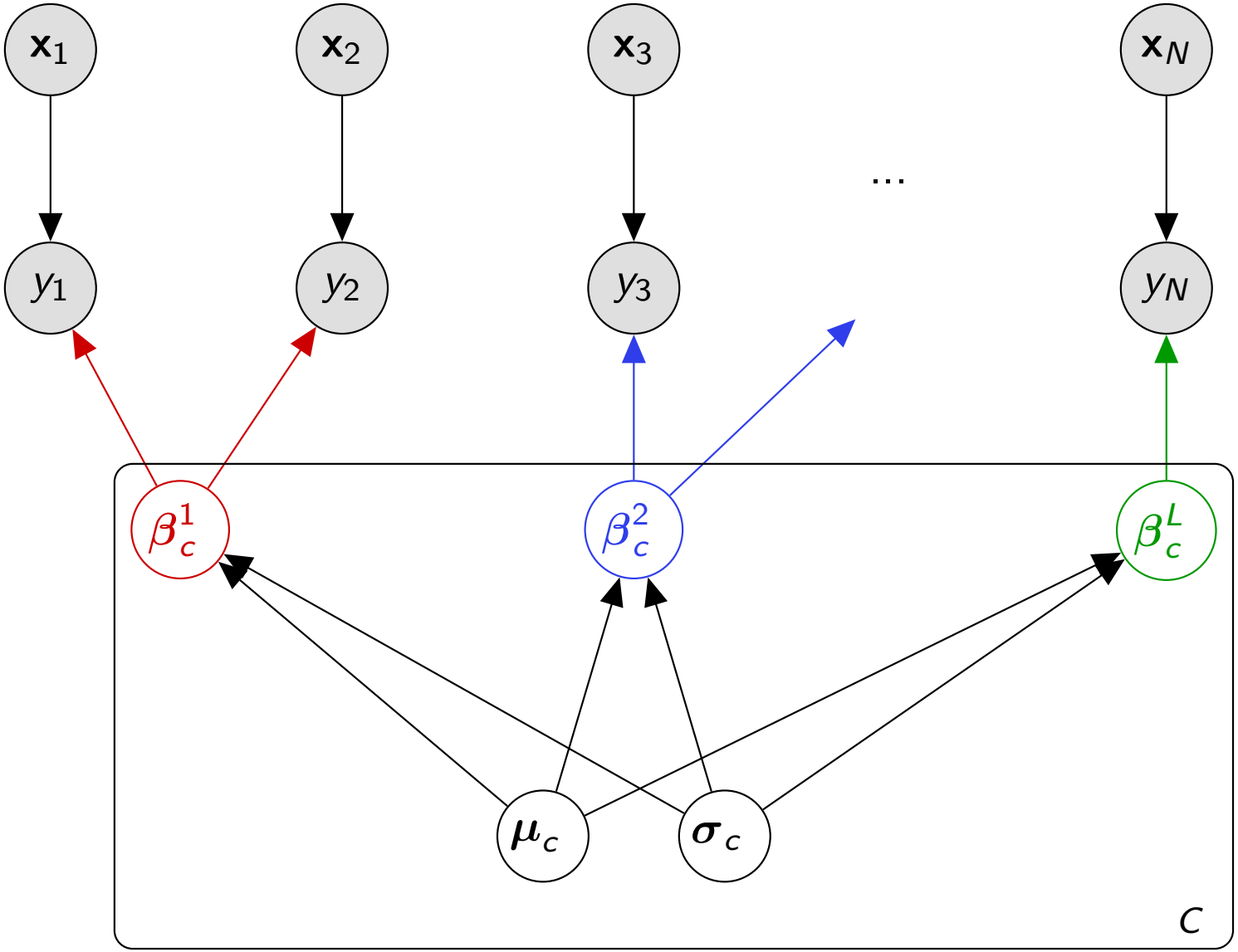


- Each individual $l \in \{1, \ldots, L\}$ gets his/her own set of parameters $\{\beta_1^l, \ldots, \beta_C^l\}$
  - Allows to capture personalized preferences and biases
  - But can lead to terrible overfitting! (more parameters than observations)

Going back to our travel mode choice case study...

- A compromise between the two: **hierarchical models**



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- A compromise between two extremes:
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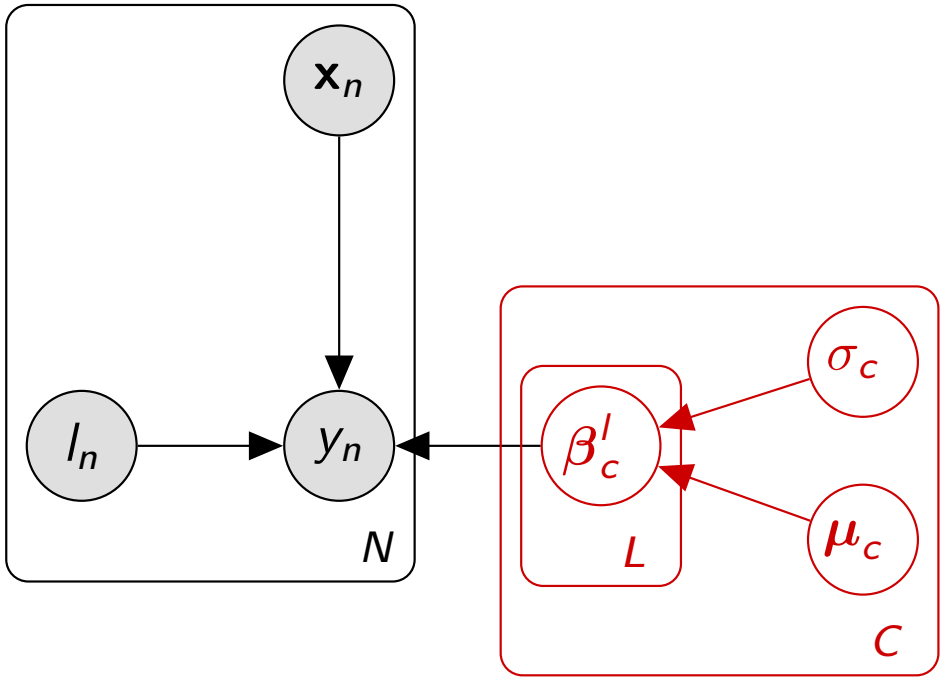
Note

This concept can also be applied to other types of models! E.g. linear regression, poisson regression, etc.

Hierarchical logistic regression model

- Probabilistic graphical model
- l_n is used to denote the level (or group) that the nth observation belongs to



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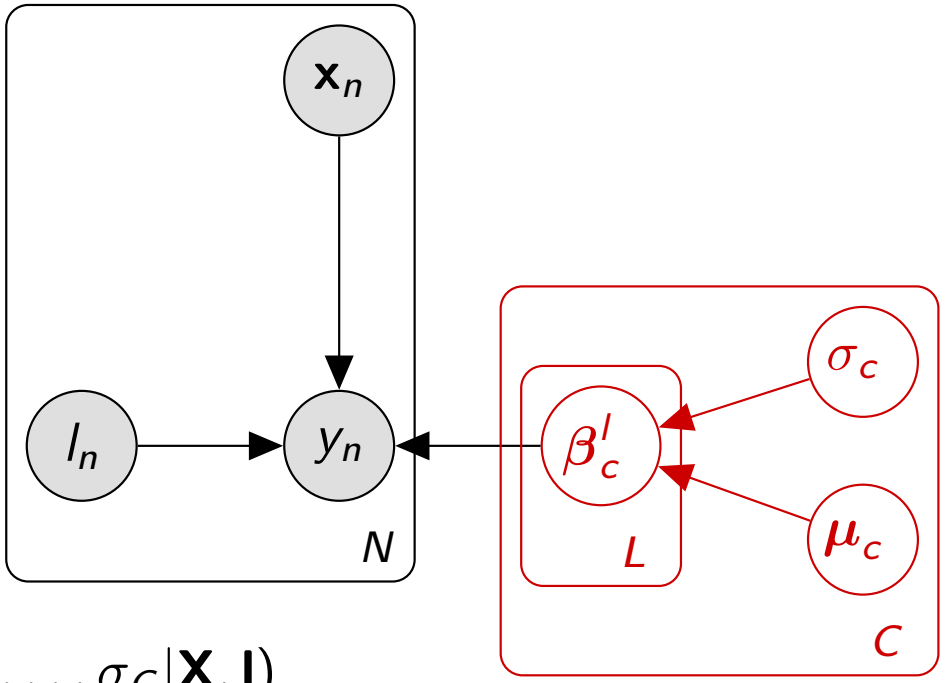

- Joint probability distribution:

$$p(\mathbf{y},\mathbf{B}^1,\ldots,\mathbf{B}^L,\boldsymbol{\mu}_1,\ldots,\boldsymbol{\mu}_C,\sigma_1,\ldots,\sigma_C|\mathbf{X},\mathbf{l})$$
$$= \underbrace{\left(\prod_{c=1}^{C} p(\boldsymbol{\mu}_c)\, p(\sigma_c) \prod_{l=1}^{L} p(\beta_c^l|\boldsymbol{\mu}_c,\sigma_c)\right)}_{\text{hierarchical prior}} \times \underbrace{\prod_{n=1}^{N} p(y_n|\mathbf{x}_n, l_n, \mathbf{B}^1,\ldots,\mathbf{B}^L)}_{\text{likelihood}}$$

where we defined $\mathbf{B}^l = \{\beta_1^l,\ldots,\beta_C^l\}$

Hierarchical logistic regression model

- Generative process

1. For each class $c \in \{1, \ldots, C\}$
   a. Draw global mean parameters $\boldsymbol{\mu}_c \sim \mathcal{N}(\boldsymbol{\mu}_c | \mathbf{0}, \lambda \mathbf{I})$
   b. Draw global variance parameter $\sigma_c \sim \mathcal{N}(\sigma_c | 0, \tau)$
   c. For each level $l \in \{1, \ldots, L\}$
      a. Draw coefficients $\beta_c^l \sim \mathcal{N}(\beta_c^l | \boldsymbol{\mu}_c, e^{\sigma_c} \mathbf{I})$
2. For each feature vector $\mathbf{x}_n$
   a. Draw class $y_n \sim \text{Categorical}(y_n | \text{Softmax}(\mathbf{x}_n, \beta_1^{l_n}, \ldots, \beta_C^{l_n}))$

Hierarchical logistic regression model

- Generative process

1. For each class $c \in \{1, \ldots, C\}$
   a. Draw global mean parameters $\boldsymbol{\mu}_c \sim \mathcal{N}(\boldsymbol{\mu}_c | \mathbf{0}, \lambda \mathbf{I})$
   b. Draw global variance parameter $\sigma_c \sim \mathcal{N}(\sigma_c | 0, \tau)$
   c. For each level $l \in \{1, \ldots, L\}$
      a. Draw coefficients $\beta_c^l \sim \mathcal{N}(\beta_c^l | \boldsymbol{\mu}_c, e^{\sigma_c} \mathbf{I})$
2. For each feature vector $\mathbf{x}_n$
   a. Draw class $y_n \sim \text{Categorical}(y_n | \text{Softmax}(\mathbf{x}_n, \beta_1^{l_n}, \ldots, \beta_C^{l_n}))$

- There are many variants of this that we can consider
  - A vector of variances $\boldsymbol{\sigma}_c$ rather than a single variance σ_c for all the features
  - Different prior distributions on $\boldsymbol{\mu}_c$, σ_c and even β_c^l
  - Hierarchical prior only on the biases (intercepts) rather than on all the β_c
  - More levels, etc.

Playtime!

- Bayesian **hierarchical** multi-class logistic regression model of travel mode choices
- Each individual has his/her own bias towards certain travel modes
- See “05 - Classification models - Part 3.ipynb" notebook
- Expected duration: 1 hour

Choice models

- An important subset of classification models are **choice models**
- These models focus on the **behavioural processes** leading to the observation of specific class labels, leveraging domain knowledge from economics and psychology
- Explaining and predicting choice behaviour is at the core of understanding human response
  - **Transport:** Mode, destination, itinerary
  - **Energy:** Appliances, energy saving measures
  - **Health:** Treatment, doctor, vaccine
  - **Marketing:** Store, brand, product, packaging

Random utility models

- **Random utility theory** is the most common theory motivating the formulation of choice models
- It posits that a decision-maker n chooses the alternative with the highest random utility out of a set of J mutually exclusive alternatives
- Under this assumption, the probability that a decision-maker chooses alternative i over any other alternative $j \neq i$ is

$$P(y_n = i) = P(U_{ni} > U_{nj} \; \forall \, j \neq i)$$

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  where U_{ni} is the utility of the alternative i for decision-maker n
- We assume that the random utility of an alternative U_{ni} is additively separable into a deterministic utility component V_{ni} and random disturbance ε_{ni}
- This implies

$$\begin{aligned} P(U_{ni} > U_{nj} \; \forall \, j \neq i) &= P(V_{ni} + \varepsilon_{ni} > V_{nj} + \varepsilon_{nj} \; \forall \, j \neq i) \\ &= P(V_{ni} - V_{nj} > \varepsilon_{nj} - \varepsilon_{ni} \; \forall \, j \neq i) \end{aligned}$$

Logit model

- We need to specify the distribution of the random disturbances
- If we assume that each random disturbance ε_{ni} is iid (independent and identically distributed) Gumbel(0,1), we obtain the **logit model**
- Under the logit model, the choice probability is

$$P(y_n = i) = \frac{\exp(V_{ni})}{\sum_j \exp(V_{nj})}$$

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- This is the logistic softmax we introduced earlier! However, the logits for the different "classes" (choice alternatives) are defined by their respective utilities V_{ni} rather than necessarily by $\boldsymbol{\beta}_i^\mathsf{T}\mathbf{x}_n$
- The utilities $V_{ni} = V(\mathbf{x}_{nj})$ can be arbitrary functions (typically linear in parameters) of the alternative characteristics $\mathbf{x}_{nj}$ ("features")
- Mode choice example:
  $V_{\text{car}} = \beta_0 + \beta_{\text{cost}} * \text{cost}_{\text{car}} + \beta_{\text{traveltime}} * \text{traveltime}_{\text{car}} + \ldots$

Identification of choice models: Only differences in utility matter I

- ML researchers are primarily interested in prediction.
  Model interpretability is secondary (though, this is changing)
- We say that a model is **identifiable** if we can estimate all parameters uniquely from data
- An unidentified model has infinitely many solutions
- In random utility models, **only differences in utility matter**
  - Suppose $J = 2$ and $V_{n1} = \beta^\top \boldsymbol{X}_{n1} + \alpha_1$, $V_{n2} = \beta^\top \boldsymbol{X}_{n2} + \alpha_2$.
    Then, $P(V_{n1} + \varepsilon_{n1} > V_{n2} + \varepsilon_{n2}) = P(V_{n1} - V_{n2} > \varepsilon_{n2} - \varepsilon_{n1})$
  - α_1, α_2 can take any value as long as $\alpha_1 - \alpha_2$ is the same

Identification of choice models: Only differences in utility matter II

- For logit, this implies the following:
  - Only $J-1$ alternative-specific constants are identified
  - Socio-demographic predictors can only enter the utilities of $J-1$ alternatives
  - Parameters pertaining to socio-demographic attributes should be alternative-specific
  - For alternative-specific variables (e.g. travel time, travel cost), we can include either generic or alternative-specific parameters

Ordinal classification

- In classification, the dependent variable is **discrete** - we predict labels of classes
- So far, we have focussed on **categorical** classification: mode choice, brand choice, blood type, cat/dog, ...
- Another type of classification problem concerns **ordinal/ordered** data: ratings (strongly disagree, disagree, neutral, agree, strongly agree), injury severity (none, light, severe, fatal), number of cars in household (0, 1, 2, more than 2), etc.

Example: Modelling injury severity

- Traffic accident reports typically contain information about the severity of the injury sustained by each individual involved in the accident
- Injury severity y_n is reported on an ordinal scale with levels "no injury", "light injury", "severe injury", "fatal injury"
- We may have access to various explanatory variables $\mathbf{x}_n$ - e.g., time of day, vehicles involved, type of road, road geometry, etc.

Ordered logit I

- We assume that there is a latent process determining injury severity
- Introduce latent variable z_n that is related to the observed injury severity as follows:

$$y_n = \begin{cases} \text{no injury} & \text{if } \kappa_0 < z_n \leq \kappa_1 \\ \text{light injury} & \text{if } \kappa_1 < z_n \leq \kappa_2 \\ \text{severe injury} & \text{if } \kappa_2 < z_n \leq \kappa_3 \\ \text{fatal injury} & \text{if } \kappa_3 < z_n \leq \kappa_4 \end{cases}$$

- κ_l with $l = 0, \ldots, l, \ldots, L$ are ordered cut-off parameters
- For identification, set $\kappa_0 = -\infty$ and $\kappa_L = \infty$ (no constant is needed because the effect is absorbed by the cut-offs)
- Then,

$$P(y_n = l) = P(\kappa_{l-1} < z_n \leq \kappa_l)$$

Ordered logit II

- We assume that z_n is additively separable into a deterministic component $V(\mathbf{x}_n)$ and a random component ε_n:

$$z_n = V(\mathbf{x}_n) + \varepsilon_n$$

- $V(\mathbf{x}_n)$ is typically assumed to be linear-in-parameters, but it can also be a neural network or a Gaussian process!
- If we assume that ε_n is logistically distributed, then we obtain the **ordered logit model**:

$$\begin{aligned} P(y_n = l) &= P(\kappa_{l-1} < V(\mathbf{x}_n) \leq \kappa_l) \\ &= \frac{1}{1 + \exp(-\kappa_l + V(\mathbf{x}_n))} - \frac{1}{1 + \exp(-\kappa_{l-1} + V(\mathbf{x}_n))} \end{aligned}$$